Chapter 8
Human resource management and software engineering

Whereas the previous chapter was intended to give the software engineer some knowledge of employee relations in the wider economy, attention will now be directed at software workers and their management. The theme is the importance of human resource management in software development and production. Criticism that the term does not accurately reflect prevailing reality has already been noted. For present purposes, this is beside the point. We might as well accept that, in the broadest sense, human resource management is a managerial crusade that, with difficulty, might be challenged on the grounds that it equates profitability or value for money with social desirability. The prescriptive and procedural elements of human resource management do, nevertheless, provide a model framework for the development of sound management practices. That is to say, whatever might be the reality of management practices in industries that are competing by redundancy and rationalization, its ostensible emphasis on the management of people, staff training and development, a strategic approach and the notion of the "learning organization" do suggest that human resource management is particularly appropriate for software work. To what extent this is so is the main theme of this chapter.

It is considered in four parts. First, a stylized, ideal type or stereotypical model of human resource management is sketched, emphasizing features salient for management of software work. Since it is recognized that such stereotypical models give an impression of stability, uniformity and tidiness that just does not exist, the second section refers to more fragmented, discontinuous aspects of software work. It is so heterogeneous that it is not an industry as such and there are growing numbers of contract and teleworkers. Nevertheless, the very phrase "software engineering" does imply a standardized approach and the considerable evidence about its costs and benefits is discussed in the third section. By way of conclusion, it is recognized that human resource management is necessarily about control and adaptability but not at the cost of proper procedures for the management of personnel.

8.1 A model of human resource management-salient features for software engineering

There is broad agreement about what constitutes the stereotypical features of a model human resource management framework. Above all, it appears to denote a strategic, proactive stance. A corresponding commitment to the organization is expected from employees. They are therefore autonomous in the sense of, to some degree, managing themselves. So a united or unitary perspective is apt because everybody can be relied

upon to pull in the same direction [1] , even though individualized for performance appraisal and pay. Nevertheless, job roles are flexible, adaptable and empowered, not rigidly and formally defined from the top. Rather than being a specialist and somewhat Cinderella function of management, human resource management is the responsibility of all managers and integrated into line management. Its final generalized feature rather begs the question. That is the maximum utilization of human resources available to the enterprise. The question is how. It is not clear that it is best achieved by meticulous human asset accounting. This is such a key issue that it is discussed in the final section of this chapter. The other features of the human resource management model and their salience for software work are now expanded in turn.

8.1.1 Long-term, strategic and proactive in style

According to Piganiol [2]

> in this new approach, the human factor is explicitly included from the very start; job analysis in the broadest sense is one of the bases for scenario-building. The scenario evaluation stage takes account of what the workers will be likely to accept, proposals they might oppose, required periods of notice and the expected training arrangements.

Human resource planning or what used to be called manpower planning is of great significance for managers responsible for locating and developing resources for the information technology environment. As Westerman & Donoghue [3] point out, the problems associated with personnel in an information technology environment require a disciplined approach to establishing numbers of staff; the utilization of personnel; the development and education of employees, together with the construction of comprehensive human resource management policies that are not only responsive to immediate

1. Fox, Man mismanagement, p. 135.
2. "Industrial relations and enterprise restructuring in France", p. 625.
3. Westerman & Donoghue, *Managing the human resource,* p. 13.

needs but also are building blocks for the medium- and long-term corporate requirements. Managers may want to make a projection of future staff needs based on expected demand and output and the necessary inputs of capital and labour. The future is irremediably unforeseeable but even a crude projection, taking into account retirements, promotions and average levels of job quits, is better than simply reacting to events. Computerized personnel management systems, including databases of employee characteristics, are a valuable tool for managers in assessing present and future staffing requirements. Such information systems go hand-in-hand with rigorous and meticulous procedures for recruitment, reward and promotion. A key issue for human resource management is skills retention and development. Although it is a fallacy that planning is impossible when change is rapid and unpredictable, we do need to take seriously the objection to excessive presumed rationality and top-down strategy:

> Arguments for the efficacy of formal planning and strategy as a method of ensuring organizational survival and growth rest upon some dubious assumptions about people's abilities to process information and, in general, to learn from experience [4] .

There has also been a growing realization that high-powered software engineering technology is next to useless unless there is a certain level of sophistication in the development environment. Thinking about software maturity assessment, some of the issues under relevant criteria are not to do with technology at all but ergonomics, such as investment in work stations or terminals; individual or open plan offices.

In any case, the integration with corporate strategy that is held to be a key feature of human resource management needs to be much broader in scope in software work. It needs, for example, to face the human resource problems encountered in trying to integrate software engineering with custom-built data-processing applications. Because users are unable to visualize the detail of a transformed working environment, it is impossible to define data processing requirements in advance. The main problem is to define the application, rather than produce well-engineered software. "Software engineering has a role to play in data processing. CASE tools produce robust applications quickly. Formal methods ensure programs consistent with applications specifications" [5] . Factory methods—further discussed in Section 8.3—may be appropriate but "correct specification of a new application has to be founded on the quaking bog of previous applications if old applications are not to become barriers to new business opportunities. The nearer software engineer

4. Staber & Aldrich, "Human resource strategies: some ecological cracks?", p. 112.
5. Cornes, "The siren song of software".

ing moves to the business definition, the more likely it is to contribute to solving the major problems of data processing" [6] .

There is no automatic linkage between advanced technology and flexibility. Some forms of software can be complex and rigid, so flexibility is a matter of design choices in the software. This, in turn, has implications for human resource strategy in software development. In short, human resource planning faces particularly severe commercial pressures in the information technology environment. It is not therefore a panacea but it can make a contribution. Westerman & Donoghue (1989) suggested several problems in software development that are more likely to occur in the absence of human resource planning: the members of a prestigious project team who have no idea of their long-term future and who are forced to leave if they are to avoid stagnation and a long-term maintenance role; the highly prized experts on one piece of software who, in a short space of time, become experts on an antique software package; the valued and loyal programmers condemned to support old undocumented systems while contractors work on new glamour applications because nobody planned the manning aspects for the total life cycle of the product; the end users who see the system as an electronic version of their previous operation who cannot appreciate the fundamental conceptual difference between processes and the power of information technology because nobody planned

their education.

8.1.2 Commitment to the organization

Originally, Walton [7] tried to define human resource management by making a distinction between control through reliance on the rules governing employment (such as in the collective bargaining model of industrial relations discussed in the previous chapter) and commitment that would come from empowering employees with status and responsibility for quality production. However, Anthony [8] was surely correct to argue that control is an ever-present requirement for management. Consequently, the real challenge is to shift employee attitudes from mere compliance with rules at work to commitment and self-motivation. Broadly defined, this signifies a commitment to staff development as part of the "learning organization" and firm-specific skills that are less transferable between firms. The new notion of skill is at once less definite and yet more vital, referring "not so much to the technical qualifications of employees but to their qualities in terms of attendance, flexibility, responsibility, discipline, identification with the company and, crucially, work-rate" [9] .

6. *Ibid.*
7. Walton, "From control to commitment in the workplace".
8. Anthony, *The foundations of management.*
9. Morgan & Sayer, *Microcircuits of capital,* p. 167.

8.1.3 Self-management

Advocates of human resource management often talk of empowering the employees with responsibility for production quality. Workers are called upon actively to exercise joint "proprietorship" of that small business denoted by their project teams and to participate in the full range of tasks involved in building a perfect product on time. This is far from total self-management, rather a kind of bounded or quasi-autonomy. Team working is a vital element. Bill Gates, among others, has made the point that software is much better done in very small teams. Some products like OS/2 break down to smaller parts that different teams can make but many products cannot be so divided. For example, Microsoft had 15 people working on the Excel product, trying to crack the original Lotus spreadsheet program. A huge amount of effort may be devoted to directly communicating with employees, whatever their grade. Since many professional jobs are decidedly autonomous, authority relations between management and worker are somewhat blurred. Direct and regular face-to-face contact between managers and workers is emphasized. As well as improving supervision, this builds trust and helps maintain motivation. The trick is to reconcile motivating individuals with team-building because it is teams, not individuals, who complete projects. This is far from straightforward because performance appraisal, assumed to be central in most human resource management strategies, can have a disastrous effect on team-building. One answer is to build it into performance management. In a case study of self-managed teams in the Digital Equipment Corporation at Colorado Springs, USA, it was found that team-based appraisal seemed to

improve participation, commitment and productivity [10] .

8.1.4 Unitary perspective

In the unitary perspective, the entire enterprise is regarded as analogous to a team with one focus of loyalty and one focus of authority. This is distinct from actual teamwork among small groups of workers. It draws a sharp contrast with the pluralistic perspective of the collective bargaining model wherein the enterprise is seen as a coalition of different interest groups [11] . Human resource management generally disparages collective bargaining of pay with groups of workers as monolithic and demotivating, unsuited to the individualistic emphasis on flexibility and commitment—but individualized pay systems can themselves be demotivating. Managements in general do not like the trade union predilection of the "rate for the job". However, when it comes

10. Norman & Zwachi, "Team appraisals—team approach".
11. Fox, *Man mismanagement,* p. 135.

to motivation, individualized performance-related pay is very much a double-edged weapon. In fact nearly all studies of motivation, whilst acknowledging that pay is important, controvert the notion that it is an effective motivator. Research by Bevan & Thompson [12] of the Institute of Manpower Studies found no link between improved company performance and performance-related pay. From their survey of Inland Revenue staff, Marsden & Richardson [13] concluded that "the positive motivational effects of performance pay had been, at most, very modest". Research conducted on behalf of the Institute of Personnel Management and the National Economic Development Office [14] cast doubt on performance-related pay as a motivator. In short, performance-related pay rarely achieves completely what it is meant to do. Despite its widespread popularity, there are often far more effective systems of motivation and reward for managements to introduce if they want to improve their organization's performance [15] .

In software development particularly, a crucial part of keeping effective workers content is a system where they can be promoted without having to become managers. At Microsoft a talented software developer can stay just that and yet rise to the top tier of elite "architects". These architects are not company directors despite their seniority, but report to the chief executive on an informal basis.

8.1.5 Flexible work roles

The human resource management approach tends to criticize bureaucracy; frankly, this is a bit of an Aunt Sally as all organizations are bureaucratic to some degree. All the same, the idea is that rather than the organization being mechanistic and centralized, with job roles formally defined, the roles are flexible with devolution of decision-making and an organic or fluid organizational structure. According to the American management consultant, Rosabeth Kanter [16] , high-tech firms in particular exhibit four main characteristics affecting their management structure: the aforementioned need to motivate

and retain technical talent; decentralization; looser authority structures and matrix organization. Conventionally, the management at a particular site or establishment is grouped according to staff function (horizontal grouping into production, marketing, accounting etc.) with delegation via line management down to first-line supervision. The matrix concept attempts to reconcile the advantages of specialization with those of co-operation across

12. Bevan & Thompson, "Performance management at the cross-roads".
13. Marsden & Richardson, *Does performance pay motivate? A study of Inland Revenue staff*.
14. Cannell & Wood, *Incentive pay: impact and evolution*.
15. Ridley, *Motivating and rewarding employees: some aspects of theory and practice*.
16. Kanter, "Variations in managerial career structures in high technology firms".

the functions and "refers to a situation where a manager or professional has reporting relationships in two directions, or, in effect, along two dimensions of a matrix" [17] . So it is also an attempt to decentralize decision-making while still maintaining the advantages of coherent central direction. For example, an engineering manager might report to both a director of engineering and a program manager for a current program or activity assignment. Matrix organization is complicated and may cause confusion, so companies that don't need it are well advised to steer clear of it.

Kanter [18] extended her argument, claiming that "the power to shape the parameters of the product and to interpret the customer's demands can be devolved extensively through corporate hierarchies. To accommodate this shift in the locus of responsibility, hierarchies may have to splinter into loose federations of product- and customer-focused units". Although change is all-pervasive and the shift to new fluid structures fairly obvious, organization theorists have struggled to conceptualize it. Paradoxically, so fissiparous are the tendencies that generalization is difficult. One attempt is the *adhocracy,* claimed to be the only structure "generally responsive to the turbulent, information intensive complexity of our changing industrial environments. It has an informal, organic structure in which people are co-ordinated by mutual adjustment and the need to create collaboratively. Authority shifts between persons whose knowledge is the most relevant at the moment and management has more to do with the integration and co-ordination of expertise than with supervision or control" [19] .

8.1.6 Integrated into line management

Rather than the personnel function being specialist and apart, human resource management is envisaged as largely integrated into line management, such that all managers are responsible for it. Writing about the "new corporate identity" and appointment of a human resource director at BA, Hopfle remarked that "this heralded a fundamental change in what had previously been the personnel department. Administrative procedures formerly attached to the personnel role were handed over to line management" [20] . It has been suggested that such developments are bad news for the future of the personnel management function. Beaumont reckoned that "perhaps the worst case scenario is that personnel will remain responsible for collective employment

issues (i.e. collective bargaining) and that line man-

17. *Ibid,* p. 113.
18. Kanter, *When giants learn to dance: mastering the challenges of strategy, management and careers in the 1990s,* p. 280.
19. Hampden Turner, "Henry Mintzberg: a profile".
20. Hopfle, "Culture and commitment: British Airways", p. 118.

agers, together with external consultants, will concentrate on the growing package of measures for the individual employee" [21] . Whilst this does not appear to have happened, the degree of integration of any genuinely strategic human resource management into line management has surely been limited in most industries.

8.1.7 Maximum utilization of human resources

Of course, improving the utilization and cost effectiveness of human resources is important but measuring information technology productivity is far from straightforward. The work processes are complex and just what ought to be the definable unit of output against which an input of human resource can be compared is controversial. Some approaches to measurement may potentially mislead managers so that they fall into the trap of concentrating on the detailed activities, rather than value-added. The following example is from hardware but also applies to software:

> ICL's Ashton-under-Lyne factory invested heavily in training and developing its workforce in order to improve flexibility and in raising standards through empowerment and quality initiatives. This shifted the emphasis from how much an hour workers are paid to how value can be added in each hour that people are employed.
>
> The approach has resulted in sharp improvements in quality, cycle time and productivity. From start to finish, a standard PC is built in ten minutes.
>
> Employee relations have also improved; labour turnover is only 1 per cent. Building on this, each member of the workforce receives a manda tory eight days of training a year [22] .

It would be misleading to suggest that such approaches are widespread.

> Typically, British managers do not see their role as being crucial in the training and development of the employees with whom they work. The prime signal managers receive from the top (for example, in the form of appraisal criteria, reward systems and promotion patterns) is that what really counts is delivering short-term results in physical and (especially) financial terms [23] .

The under-side of human resource management, for which decentralization and empowerment may provide a façade, is its potential, particularly

21. Beaumont, *Human resource management,* p. 13.

22. Wheatley, "Revival begins at home for computer makers".
23. Storey & Sisson, *Managing human resources and industrial relations,* p. 176.

when combined with information technology, for intrusive surveillance of employees. According to Winfield effective use of human resources now means exactly what it says:

> It is the possibility of computer surveillance of work rate that allows decision makers to look more critically now than ever before at work output in offices. Simply attending—selling your time—may have been tolerated in the past; what is now possible and indeed demanded by monitoring and control systems is nothing less than effective work. It is nothing short of people's attention and total dedication to the task in hand that is now being asked for [24] .

At a London conference on electronics and privacy in June 1999 it was reported that all staff using computers should now assume that they are being monitored. Research by Infosecurity, a computer security concern, found that office workers spend an average of 30 minutes a day using on-line facilities, such as buying and selling shares, rather than working for their employers. In June 1999 a woman lost an unfair dismissal case after being fired for booking a holiday on the Internet from her workplace after 150 web searches [25] . Since it pertains to the contract of employment, this seems to be an issue that will give rise to great legal creativity with various defences available. A secretary who was sacked because of her "Internet addiction" claimed unfair dismissal in the first case of its kind to go before an employment tribunal in August 1999. If the tribunal had found that she had an addiction, her lawyers would have been able to make a claim under the Disability and Discrimination Act 1995 that provides for unlimited compensation [26] . In October 1999 a senior female executive was dismissed after she sent an e-mail to her firm's personnel department expressing disgust at the homosexual practices of another employee.

Human resource management is therefore Janus-faced. It promises participation but may deliver continuous monitoring and surveillance. This is some way from the high trust, commitment model that some advocates of human resource management as something completely different try to portray. However, the human resource management commitment model is optimal for software work. This is because programmers must keep drawing attention to their failures and be rewarded for doing so. "This is a tough proposition for commercial companies and almost unthinkable in adversarial (them or us) or fear-based cultures. Management gets the impression that the

24. Winfield, Organisations and information technology, p. 30.
25. Sabbagh, "'Staid' Britons share the online profits".
26. Verraik, "Woman fired for internet over-use".

project is going well and has no idea what's actually happening at the grass roots level. By the time they find out, it's too late" [27] .

8.2 The structure of software development and production

Software development and production in the sense of manufacturing can be distinguished. The main business is software development, with mass production of software packages sometimes integrated, sometimes sub-contracted. While full-time employees form the majority of workers in software development and production, the business is so heterogeneous and fragmented that to speak of an industry is potentially misleading. Many software workers are employed by companies whose software development is definitely ancillary to their main business. They may not be interested in commercializing their own software in case this distracts from delivering systems for the main business.

Witham [28] provided a summary of the various categories of non-permanently employed workers. A freelance worker is one to whom work normally done by a permanent employee is sub-contracted. The freelance contacts the end-user, arranges the work content and duration, scale of charges and schedule directly, signs a contract directly with the end-user and performs the work either on-site or at base. Freelancing also signifies that the sub-contractor is an individual without partners or employees.

Body-shopping is not as such a method of working but, more accurately, refers to the relationship between the worker and employer. The individual is selected to perform a piece of work by virtue of relevant experience and availability. The immediate employer—agency, consultancy or software house—will have no say in the content of the job which is determined by the end-user of the contract services. Agency probably means freelance agency where the agency is responsible for finding the work and arranging contractual details with the client. A variant is employment by agency where the contract between freelancer and agency is an employment contract, usually for only the duration of the work stipulated by the client. When an agency deploys complete project teams, it is known as a contract house. A contract house is different from a software house in that self-employed people are used and in not itself specifying the nature of the work or its technical content, just recruiting and paying the workers. Software houses may use some self-employed workers and also supply their own body-shopped workers. The human resource plan of a good software house may be to do as much work as possible either in-house or, if on-site, under their own human resource management. Even so, the software house may be obliged to body-shop its

27. Schofield, "Counting the cost of failure.
28. Witham. *"Computer guide to the contract industry"*.

own workers. Workers earning fees are better than idle hands. The software house gains a contact and possible future client. Sometimes work for which the software house originally tendered has been taken over by the client but the original software project team stays on-site. Witham commented that the better software houses do not like this practice but more rapacious outfits that purport to be software houses but are really glorified body-shops, are less fastidious.

While the self-employed are a fundamental resource in information technology

contract labour, Witham reckoned that there are relatively few on account of financial obstacles presented by current legislation. Whether or not that is the case—and it is debatable—software is especially suited for entrepreneurial, sub-contracted networks. Norman Macrae, formerly of the *Economist* was among the first to predict such globalized entrepreneurial commerce, with most work accomplished by free agents using computer networks. The prototype version of this is teleworking by homeworkers or employees of a business or government agency who work part-time or full-time at home rather than at the employer's premises. The production technology involves some mix of office computer system, home PC and telephone line.

Teleworking is simply work by remote control. Usually the worker is working from home and maintaining contact with the employer or, if self-employed, other business, through communications networks. It reduces commuting by making the work flow to and from the worker. It could have a significant impact on public transport systems in London and other big cities. Partly for this reason the UK government published a guide to teleworking and supports this method of flexible employment.

Because teleworking means different things to different people, it is hard to quantify how many people are involved. The expression can be used to mean anything from keying in data at a remote telecentre factory—such as centres used by the major commercial banks—to travelling sales staff with laptop computers and the idyllic life of the well-paid professional working from a country cottage, yet maintaining employee status and a desk at head office with the individual bargaining power of scarce skills.

In fact, for many, the reality of working at least part of the time from home, linked to the office with a computer and a modem, is rather less idyllic. Teleworking is often little more than a hi-tech version of traditional industrial homeworking—with the employer saving overhead and child care costs. Not surprisingly, telecommunications and IT companies, such as BT, IBM and Digital, have led this change in working practices. While BT have offered employees home-based telework as a union-negotiated option, others, including sections of the Civil Service, British Gas and Scottish Power, have introduced telework to cut costs and increase the commercial advantages of flexible working. Ideally, employer and employee gain: for example, when the Co-operative Bank gave employees in its bad debt collection department the chance to work from home, productivity went up to 40 per cent—an example, perhaps, of the commitment model of human resource management.

Trade unions tend to be suspicious of all kinds of homeworking. It is a form of employment that makes collective organization and representation very difficult. At least the Communication Workers' Union is in a good position to wire up local union offices. However, BIFU, the Banking Insurance and Finance Union, has found cases of members compelled to work from home on a lower grade.

Inevitable as the growth of teleworking will be, it must be acknowledged that it is not suitable for everyone. Furthermore, whilst it has been associated with a shorter working week, this is slow in coming. The greatest challenge is for managers to manage people whom they rarely see. All the talk of corporate culture is difficult to reconcile with a widely dispersed workforce less able to feel genuinely part of the company.

There are many examples of "off-shore"-telework. American firms use local workers for electronically linked data processing and administrative work in countries such as

Barbados, China, India, Ireland, Jamaica, Korea and Mexico. This is only one aspect of the globalization affecting software production. Some consider that programming is too lowly a task to be done in the UK or Singapore and is more efficiently sub-contracted to India or China where intelligent and experienced graduate programmers will do it for a lower price. Ed Yourdon [29] predicted massive unemployment among the ranks of American programmers, systems analysts and software engineers, warning that international competition will put American programmers out of work, just as Japanese competition put American automobile workers out of work. Yourdon's solution was building world class software development teams. In fact, the US software industry was well ahead of the Japanese. Japan had constructed literal software factories and it was argued [30] that these posed a challenge to Western management. The failure of the challenge is discussed next.

8.3 The software factory

The analogy of software production with industrial scientific management principles goes back to the research of Kraft & Dubnoff [31] . It is a rather different issue to posit that standardized techniques can be applied to software development, as though it were a type of manufacturing. In general, it is inaccurate to conflate the term software engineering with rigid division of labour. It is more to do with structured programming, i.e. a logical, struc

29. Yourdon, *Decline and fall of the American programmer.*
30. Cusumano, *Japan's software factories: a challenge to US management.*
31. Kraft & Dubnoff, "Job content, fragmentation and control in computer software work"

tured approach to programming and with software tools to assist productivity. An early definition of software engineering was "the establishment and use of sound engineering principles in order to obtain software that is reliable and works efficiently on real machines" [32] . By setting out the three key elements of methods, tools and procedures, it enabled the manager to control the process of software development. The methods refer to the wide range of tasks in building software—project planning and estimation; system and software requirements analysis; design of data structure; program architecture and algorithm procedure; coding; testing and maintenance. The methods often introduce language-oriented notation and criteria for software quality. Tools provide automated support for methods. They may be integrated so that the information created by one tool can be used by another, thereby establishing a system for the support of software development (CASE). The procedures hold the methods and tools together and control timing by defining a sequence in which the methods will be applied.

However, creating the software that actually does the work is not completely an engineering discipline, even though some software professionals aspire to that. Computer programs are mostly still written the same way they have been for 40 years—by hand—and individual programmers are much more like creative writers than brickies. This is why large-scale government computerization projects are much more likely to go wrong

[33] .

Clearly, the aim of an industrial approach is more effective management to maintain and enhance quality and productivity. The question is whether increased structuring will achieve that or run the risk of "over-engineering" and detachment from end-user environment that may emanate from the "siren song of software" [34] . Since, although Cusumano's book is not a prescription for management, it examines this issue in detail, it is worthy of further comments. At the beginning of the 1990s, Toshiba operated the Fuchu software factory employing 2,300; Hitachi had two software establishments employing 4,000 programmers and systems engineers and 7,000 employees in subsidiaries and sub-contractors. Fujitsu had two software factories employing 3,500 engineers with 14,000 employees among subsidiaries and sub-contractors. NEC operated five software factories with just under 10,000 employees. According to Cusumano, these large scale operations had solved the classical problem in software engineering—how to replace craft control and design autonomy with factory discipline, reliable planning and systematic quality control.

The companies gained from labour in two ways: "The prospect of lifetime employment and long term careers within these large firms guaranteed that

32. Bauer, *Software engineering*.
33. Schofield, "Counting cost of failure".
34. Cornes, "The siren song of software".

programme engineers did not quit, as many would probably have done in the US" [35] . Furthermore, they could hire young workers unskilled in software but train them to use a standardized methodology and tool set as reusable designs. However, as the down-sizing and flexibilities of the 1990s accelerated, the software factories proved out-moded. Software was no longer seen—as in the Japanese *keiretsu* model—as an addition to hardware. Rather, "the *keiretsu* structure had inhibited the development of general standards and strong software vendors and, with standards now emerging, they are coming from external players, that is US firms" [36] .

Rather than pointing to a solution to the management problems that occur in software development and production, as in other business activities, Cusumano may have inadvertently encouraged readers to make a typical management mistake by seeking a process to solve a problem involving people. In that context, the challenge for human resource management is to replace compliance with rules with trust and support, to replace the idea that employees serve the manager with the idea that managers enable the employees to do their work—and to empower the employees to use their energy and creativity on behalf of the business. Reviewing Cusumano's book, Crocca suggested that "just as the Japanese adopted the scientific method, we have to adopt the emotional method: be sensitive when dealing with people and analytical when dealing with process. Management must resist looking to science for "silver bullet" solutions to people problems" [37] .

8.4 Training and human resource management

In the first edition of *Professional Issues in Software Engineering,* the software gap—or skills gap—was discussed. This has not been filled. Despite universities establishing more IT and computing courses and applications rising strongly, the industry continues to generate more vacancies than capable recruits. Research for Microsoft predicted 1.6 million job vacancies in IT by 2002 when the industry and universities increase training places. The perception of IT careers does not help in recruitment. Mark East, education group manager at Microsoft UK was reported in June 1999 as believing that "the view is that most techies are nerds with beards who program all day" [38] .

It is true that university computer science degrees tend to be pseudo-scientific and rather theoretical. More traditional courses do emphasize programming and the theory of software engineering. Consequently, students

35. Berggren, "Japan as number two".
36. *Ibid.*
37. Crocca, "Review of M. Cusumano, Japan's software factories".
38. Pritchard, "Great jobs, good money, few takers".

associate computing degrees with science and maths—both areas where applications have fallen over the last decade.

By contrast, there are innovative courses aimed at multimedia. Furthermore, from an industry point of view, the mathematically-based computing degree is not always necessary. Business management needs graduates who can work with IT without the theory of computer science. "Research carried out by NOP for Microsoft found that 20 per cent of people entering the IT industry did so without formal IT skills" [39] . Computing companies find that IT graduates often lack transferable or "people-handling" skills, such as communications and a broader knowledge of how businesses work. Dr Neil Barrett, senior fellow at Bull, reckoned that "from an industry point of view, we are often better placed to take people with good generalist degrees and turn them into engineers". Although computer scientists have learned to use elegant, compact languages and pure programming techniques, they can stumble in creating real applications in a real workplace. "They are people who understand the finer details of software programming but cannot program. We have to start again and teach them the methods and tricks we work with" [40] . The answer may be degree schemes that combine computer science with business studies. However, a more radical solution may be for the industry to create some sort of mechanism that can compensate for job insecurity, perhaps by creating the equivalent of a Dock Labour Board of permanently paid employees on whom the various software companies can draw when necessary.

8.5 Conclusion: human resource management in software—commitment and control

The tensions and contradictions between a factory approach and human resource management can be demonstrated by reference to a software package using the programming technique of object-oriented software. An objectview presumes that any transaction or business operation starts with an object and gives a choice of related activities. An object can be any item used by anyone in her everyday work. Activities to do with that object—information, an invoice, even a diary date for a customer—are all options attached to that object and a user simply picks the relevant activity.

Accessibility to information was the feature that convinced a multinational health care group's management to spend an initial $500,000 on a revamped personnel package of this type. The company had been reorganized to emphasize European, rather than local, control of operations in the region. Managers in its five business lines became responsible for profit and loss,

39. *Ibid.*
40. *Ibid.*

making information about salaries and other employment costs a vital part of the European picture. As the director of human resources put it, "we call it a human resources system but it is really a business system because line managers will have access to it" [41] . He advocated a devolution of personnel responsibilities to line managers. The application of the package was aimed at giving a thousand managers easy access to the Europe-wide information that they need to assess and react to the ever-changing deployment of staff.

The case hints at the dichotomy in human resource management. This is to say, control by rules or control by influencing the organizational culture to emphasize loyalty, seriousness, commitment and responsibility. Walton [42] tried to extract the essence of human resource management by saying that it necessitated a move "from control to commitment". But this will not really do because, as Anthony put it, "if management is not accepting responsibility for the control of labour, then it is not managing" [43] .

Rather, the contrast that human resource management displays is between compliance and commitment as different forms of managerial control. The main expression so far of human resource management is in the trend to integrate personnel management with senior and middle line management. The idea is to move away from compliance-based and rules-based systems of control that often involved collective bargaining about pay and conditions in a centralized, mechanistic organization structure. Instead, Kanter [44] has advocated forms of control that instil personal commitment because the rules and procedures "stifle initiative and creativity" in an atmosphere that is "emotionally repressive". Therefore human resource management purports to aim at "responsible autonomy" [45] , empowering employees with responsibility for production and product quality. Unfortunately, however, the main approach to this so far is a kind of smothering dogma. For example, a mission statement is introduced, there are strategic plans and

constant talk of changing the culture that implies a monolithic unitary perspective for the organization. "Commitment" seems to involve "the internalization of management-derived and sanctioned beliefs, norms and values, in the sense that they become part of the core of the individual's perceptual world; thereby they develop into moral obligations" [46] . Moreover, for all the talk of culture management and empowerment, there seem to be even more layers of bureaucratic control than under explicitly rule-based systems. Much of this

41. Gooding. "New objective for software".
42. Walton, "From control to commitment in the workplace".
43. Anthony, *The foundations of management,* p. 1.
44. Kanter, *When giants learn to dance: mastering the challenge of strategy, management and careers in the 1990s,* p. 20.
45. Friedman, *Industry and labour,* p. 84.
46. Johnson & Gill, *Management control and organizational behaviour,* p. 36.

"re-bureaucratization" is concerned with measuring outputs and performance and there is a great growth industry in inspection and monitoring.

Management may attempt to dispense with responsibility for control of some employees by use of sub-contracting. Many companies have done this for cleaning and catering services. But this is hardly a move to commitment as sub-contracted workers cannot be expected to have loyalty to anyone. So commitment can be expected only among core, permanent workers. Even here, commitment may be one-sided. Vicki Smith [47] observed that a commitment to paternalistic lifetime employment patterns by a banking company concealed a covert plan for cutting personnel—charging middle managers with "managing up" (raising individual productivity) or "managing out". The device used was the performance appraisal in order to penalize or push employees. Here, moving from compliance to commitment meant that managers should "'reorient their thinking away from bureaucratically constricting standardized job descriptions based on position and focus instead on increasing results in order to contribute to improved corporate performance". This they would do by setting "stretch" objectives: upgrading people by pushing them to achieve ever higher levels of output.

The sort of software discussed at the beginning of this section and the TARDIS example in the previous chapter show that human resource management can be a subterfuge for intensification of the bureaucratic approach and its actual centralization. "Assessing and reacting to the ever-changing deployment of staff" is facilitated by labour hours analysis, carried out on a PC and taking account of operator efficiency, attendance ratios, "non-productive work" ratios, learning curves etc. Human resource management may therefore mean an intrusive surveillance of employee performance.

Even if this dark side of human resource management is acceptable on grounds of efficiency, "business need", flexibility and commitment do not justify the manipulation and favouritism that can be unleashed by passing personnel management to line management. If moving to commitment entails extinction of necessarily bureaucratic rules of procedure and consistency, human resource management is humbug. Fair treatment of all employees, due process (for disciplinary cases etc.) and consistency must

be the foundation of any human resource management that aims to secure the trust and commitment of employees.

This is obviously the opposite of "management's right to manage irrespective of whether this is fair and reasonable" [48] of concealing information from subordinates, feigning ignorance, side-stepping of official procedures to gain one's own way and resort to expediency. Properly carried out and fairly

47. Smith, *Managing in the corporate interest,* p. 47.
48. Clark, "Procedures and consistency versus flexibility and commitment in employee relations", p. 80.

implemented, the human resource management model is about procedures in employment relations. Systematic recruitment and selection techniques, training needs analysis and performance review are fundamental features of the model. Consequently, although human resource management must recognize that managers operate within markets, if it means anything, it also means that they do so by "assembling and maintaining human associations in sustainable organizational forms" [49] . Essentially, the challenge for managers of software development is that high-trust, knowledge-based organizations need managers who can cope with complex, social, political and moral environments. Over-emphasis on the "bottom-line" and resource facet "ignores the institution-building and nurturing role of management so clearly recognized by our competitors" [50] . The importance of procedures for consistency should be apparent to software engineering managers more than most. So even if we accept that world-class software organizations will move aggressively towards specification-level reuse because the specification models are contained in CASE repositories—the people who constitute the project team cannot be stored in a repository. Nevertheless, their experience and co-partnership ought to be maintained—by human resource management approaches to software productivity which often result in bigger gains than technical approaches.

8.6 References

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